

Metal powder bed fusion.

Powder bed fusion, also known as powder bed melting, is the most used method for 3D metal printing. In this method a hopper uniformly distributes a fine layer of metal powder over the build plate and the powder is melted using an energy beam in a desired shape to form a cross-sectional layer. The hopper re-applies the metal powder and the process is repeated over and over again. The thickness of metal powder is commonly between 25µm and 100µm, but the optimal thickness of each layer of spread powder depends on the processing conditions and material used. There are two types of powder bed fusion technique:

Electron beam melting (EBM).

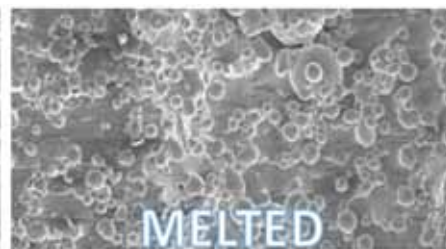
EBM is a type of metal-powder bed fusion technique where an electron beam is used for melting the metal powder. It requires higher energy compared to the other (SLM) method described below. The process takes place in a vacuumed environment. This method requires less tooling and setup cost. One of the main drawbacks of this method is that the surface finish of the printed product is poor and is similar to that of the sand casting technique.

Selective laser melting (SLM).

SLM is also a type of metal-powder bed fusion technique where a laser is used for sintering the metal powder. This method is capable of working with nearly all alloys. Furthermore, the mechanical properties of the printed parts are equal to conventionally formed parts. But this technique is relatively slow and requires expertise to achieve quality print. Also, expensive post-processing is required before printed objects are ready for use.

Direct energy deposition

(DED). Direct energy deposition is a technique of metal 3D printing which uses a metal feedstock either in the form of a metal powder or



Difference between the internal structure of melted and sintered metals (Source: www.additivallab.com)



Steps involved in the BPE method (Source: www.exbuild.com)

wire and a heat source, such as a laser beam, to fabricate parts. The laser and the feedstock sit on a single print head. The method can be divided on the basis of feedstock as:

Wire DED. Wire-directed energy deposition is a technique where a metal wire is used as a feedstock in place of metal powder. A heat source—such as an electric arc, laser, or electron beam—is used as the energy source to continuously melt and deposit the metal wire along a predetermined path. The wire DED is simpler and cheaper than power based metal 3D printing. It also has a high deposition rate, which makes the process faster. When compared to metal powder, the wire as a feedstock enables higher quality prints and higher overall safety along with lower print cost.

Bound powder extrusion (BPE).

BPE is one of the newer techniques where the feedstock is a metal powder bound together with a waxy polymer. This makes the technique much safer than one with loose powder. The BPE filament is extruded out of a nozzle into the desired shape. After printing, the object has to go through

two more steps, which include washing out and sintering the 3D object. The washing removes any traces of wax polymer and the sintering allows the part to shrink and fill any spaces opened up by the dissolved binder, thus yielding a solid metallic part.

This method is more affordable than most other methods and is easier to use. Furthermore, a wide variety of materials can be used as a feedstock and the parts can be printed with open cell infill, thus reducing part weight.

Metal binder jetting. In the metal binder jetting method, a liquid binder is used to join powdered metal particles together. This process works similar to a 2D printer, which puts jet ink onto the paper. In this method, a fine layer of metal powder is spread across the print bed and the binding polymer is sprayed in the shape of the part cross-section, thus, loosely joining the parts together. Once the 3D shape is formed, it requires post-processing, such as sintering, to solidify the part.

The metal binder jetting technique requires considerably lower post-processing as it overcomes a few of the drawbacks that are